

Placeholder

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Abstract

The emergence of Large Audio Language Models (LALMs) has broadened access to advanced and currently developing language technologies by supporting audio input alongside text. However, this convenience has primarily benefited speakers of high-resource languages, when this can be valuable to speakers of languages in low-resource settings, especially those where speech is the primary mode of communication. While recent research has focused on including low-resource languages by simply restricting training to a linear projector connecting a speech encoder and a large language model, as the decoder, the impact of the specialization of the large language model (LLM) used as the decoder remains unexplored in this setup. In this work, we investigate decoder specialization using a family of state-of-the-art (SOTA) small multilingual LLMs developed specifically to address the challenges of low-resource languages.

Extensive experiments across 12 languages, spanning a range of resource levels, including morphologically rich languages that remain challenging for automatic speech recognition (ASR) systems, reveal only minor differences in character error rate (CER) between the three regionally specialized decoders and their general-purpose counterpart. We release all checkpoints to support further investigation.

1 Introduction

Automatic speech recognition (ASR) models such as Whisper (Radford et al.) and Omnilingual ASR (team et al., 2025) have expanded multilingual speech recognition to a growing number of languages, including low-resource languages. Despite this progress, substantial work remains to address the breadth and diversity of low resource languages and dialects, which has motivated recent studies to investigate the failure modes of these systems in low-resource settings (Imam et al.; Kyslyi

et al.; Kumar et al., 2026; Bharathi et al.; Azis et al., 2026). These limitations are important to address because language technologies today are increasingly multimodal, although text-based interfaces as the default in many settings and the speech modality is still developed primarily for high-resource languages. This is crucial given that, of the roughly 7,000 languages spoken worldwide, around 4,000 are estimated to have an established writing system, however, it does not correspond to widespread literacy or everyday use among speakers¹. Considering this, Speech-first technologies may benefit speakers of languages in low resource settings by enabling them access to services that are better supported in high resource languages with current technologies. However developing end-to-end speech language systems remains computationally expensive and relies on abundant, high-quality annotated data.

One recent direction addressing these constraints is the SLAM-ASR architecture (Ma et al., 2024), which connects a pretrained speech encoder to a large language model (LLM) through a trainable linear projector, while keeping the pretrained models frozen. Restricting training to the projector provides a practical way to introduce a language in low-resource settings. Studies building on this work have investigated its robustness (Kumar et al., 2025), number of necessary audio hours (Fong et al., 2025), which encoder layers can be pruned while maintaining performance through fine-tuning of the LLM backbone (Kolluri et al., 2026), language-level and language-family-level training configurations (Zhang et al., 2026), and reducing prompt sensitivity with a learnable prompt projector (Burdizzo et al., 2026). Although these prior studies have investigated several aspects of SLAM-ASR, the decoder selection has generally been guided by target-language support and state-

¹Ethnologue, 28th edition <https://www.ethnologue.com/>

of-the-art (SOTA) multilingual coverage, [TODO: Add Examples of the LLMs they used]. To the best of our knowledge, however, the effect of decoder specialization within this architecture has not been systematically evaluated, leaving open whether specialized decoders provide an advantage over general-purpose models unexplored. More recently, the Tiny Aya family(Salamanca et al., 2026) introduced SOTA small multilingual LLMs that share a common multilingual foundation, with each regional model having undergone post-training to reinforce the linguistic clusters associated with its respective region.

Motivated by the growing study of failure modes in low-resource ASR and recent advances in small regionally specialized multilingual LLMs, we aim to contribute to the efforts to build linguistically informed language technologies. In this study we address the unexplored role of decoder specialization within the SLAM-ASR architecture by examining empirically comparing regional Tiny Aya decoders with Tiny Aya Global across 10 languages , [TODO: Update]nine are from the three Tiny Aya regional clusters and one language unsupported by both and examining whether and to what extent specialization offers a strong starting point for low-resource languages when ASR data are scarce. [TODO: Introduce research questions(s) ?]

SLAM-ASR as a route into low-resource languages. The appeal of this architecture for low-resource settings is that almost nothing has to be trained. The encoder and the decoder are frozen and can be taken off the shelf, and the only optimized component is a projector of a few million parameters, so a language can be added with a few hours of transcribed audio and a single GPU rather than with a full ASR training pipeline. Fong et al. (2025) quantify where that route becomes competitive, finding that roughly 100–200 hours are needed before the framework matches a Whisper-only baseline, and that projector pretraining on a high-resource language substantially improves the low-resource end. Reviews of African ASR report the same shortage from the data side (Imam et al., 2025; Adjovi et al., 2026).

Most SLAM-ASR studies to date evaluate on a small set of well-resourced languages. Of the twelve languages we study, seven — French, Marathi, Urdu, Swahili, Hausa, Amharic and Kreol Seselwa — are not examined in the speech-LLM literature we survey, and Kreol Seselwa appears

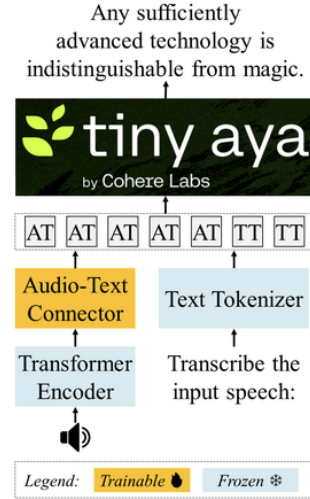


Figure 1: LisTaya

not to have been evaluated in this architecture at all. Reporting them together, under one recipe and one encoder, is part of what this study contributes.

2 Methods

2.1 SLAM-ASR

[TODO: then remove specifics of encoder and decoder names and write them at the end.] Our implementation follows the Encoder–Connector–Decoder architecture introduced in SLAM-ASR Ma et al. (2024) and is built upon the Qwen2-Audio codebase ², as Illustrated in Figure 1

Following the original SLAM-ASR framework, only the Connector (projector) is optimized during training, while both the encoder and the decoder remain frozen. This design enables efficient adaptation by learning the alignment between speech representations and the language model embedding space while minimizing the number of trainable parameters.

2.2 Decoder Specialization

PH Note: Add details regarding the Tiny Aya Family, include a table.

To study the effect of decoder specialization on multilingual ASR, we employ models from the *TinyAya* family as the frozen decoder within the SLAM-ASR architecture.

TinyAya-Base is a pretrained 3.35B-parameter multilingual language model covering more than 70 languages. Built upon this foundation, *TinyAya-*

²Qwen2-Audio codebase, [Qwen2-Audiocodebase](#)

Global provides balanced instruction-tuned performance across 67 languages, while three regionally specialized variants are available: *TinyAya-Earth* for languages from Africa and West Asia, *TinyAya-Fire* for South Asian languages, and *TinyAya-Water* for languages from the Asia-Pacific and Europe.

[TODO: Post training for each variant, tokenizer specialization]

All TinyAya variants share a common multilingual tokenizer, which is trained using a balanced weighting strategy that combines data-distribution and language-family information to achieve equitable representation of both high- and low-resource languages. Furthermore, TinyAya adopts a region-aware multilingual posttraining strategy in which high-quality instruction-tuning data is constructed through language-specific translation, prompt transformation, and Fusion-of-N teacher generation, enabling balanced multilingual instruction tuning across diverse language groups while improving coverage for low-resource languages.

3 Experimental Setup

3.1 Datasets

Our experiments use two multilingual speech corpora: World Speech (Asonitis et al., 2026) and FLEURS (Conneau et al., 2023). WorldSpeech primarily contains formal speech collected from diverse public sources such as parliamentary proceedings, international broadcasts, with smaller contributions from public-domain audiobooks. FLEURS contains read speech based on parallel sentences from the FLoRes-101 dataset, recorded were under relatively controlled conditions. We use subsets of train sets from WorldSpeech for training, excluding audio samples with durations of 30 seconds or longer. For evaluation, we use the validation sets from FLEURS when available and on the WorldSpeech test sets otherwise.

We select 12 languages for this study (Table 1). 11 are covered by the regional Tiny Aya models: Water (English, French, Spanish, and Indonesian), Fire (Hindi, Tamil, Marathi, and Urdu), and Earth (Swahili, Hausa, Amharic). Kreol Seselwa is an African language selected because it is unsupported by both backbone models. We include it to assess whether the trainable connector (projector) can effectively learn the alignment between speech representations and the language model, despite this lack of support.

Language	WorldSpeech	FLEURS
	Train	Test
English	en_us	en_us
Spanish	es_mx	es_419
French	fr_ca	fr_fr
Indonesian	in_in	id_id
Hindi	hi_in	hi_in
Marathi	mr_in	mr_in
Tamil	ta_in	ta_in
Urdu	ur_in + ur_pk	ur_pk
Swahili	sw_ke + sw_tz	sw_ke
Amharic	am_te	am_et
Hausa	ha_ng + ha_td	ha_ng
Kreol Seselwa	crs_sc	crs_sc

Table 1: Dataset overview. WorldSpeech training subsets are used to train the projector. Evaluation uses the FLEURS validation split; bold entries use the WorldSpeech test split. The + symbol denotes subsets pooled.

3.2 Training Configuration

In our study, we use **Listening Tiny Aya** (LisTaya; see Figure 1) to refer to models that use Whisper Medium as the fixed speech encoder, paired with different regional variants from the Tiny Aya family as decoders. Depending on the specialized regional decoder we refer to the resulting models as LisTaya-Water, LisTaya-Fire, LisTaya-Earth; LisTaya-Global for the general-purpose variant.

Throughout all experiments, we train only the projector while keeping the encoder and decoder components frozen. We use the AdamW optimizer with a learning rate set at 1.5e-3, training steps 1, 000, batch size 8, gradient accumulation of 64 (an effective batch size of 512), and no weight decay. We retain the checkpoint after 1,000 steps, as well as an additional checkpoint achieving the lowest validation CER. All models were trained with one NVIDIA H200 GPU.

4 Results

4.1 Decoder specialization does not change what the model hears

Table 2 reports, for each language, the best CER of the region-matched decoder against the mean of the two mismatched regional decoders and against LisTaya-Global. Across the 11 languages covered by a Tiny Aya region, the matched decoder is **0.55**

CER better on average, with a 95% confidence interval of $[-3.89, 1.99]$ and a Wilcoxon signed-rank $p = 0.966$. Six of eleven languages favour the matched decoder and five do not.

This is a null, and its width is what makes it informative. The paired design’s minimum detectable effect is **4.90 CER**, so effects smaller than that cannot be resolved here. We also measure the run-to-run floor directly rather than assuming it: over ten replicate pairs in which a cell was trained twice, the between-run standard deviation is **1.17 CER**, against a median within-run late-training standard deviation of 1.09. The observed effect is smaller than the noise between two runs of the same configuration.

4.2 The specialization is small in the data

A null invites the question of whether the manipulation was large enough to detect. We quantify it from the Tiny Aya report’s own language-composition tables: for each language we take the share of the matched variant’s post-training mix in that language, minus the mean share of the two mismatched variants. The median excess is **1.3 percentage points**.

Regional specialization in this model family is therefore a shift of about one percentage point of post-training data per language, not a change of training corpus. Read against that, the result is better stated as *specialization at this magnitude does not measurably change recognition accuracy*, which is a narrower and more useful claim than specialization being inert.

4.3 Training data volume predicts overfitting

While decoder choice does not predict accuracy, the amount of training audio predicts how a run behaves. Table 3 groups languages by resource tier and reports the rise in evaluation loss after its own minimum, and the fraction of the run elapsed at that minimum. Both are monotone across the four tiers: the lowest-resource cells reach their best evaluation loss about a quarter of the way through training and degrade over the remaining three quarters, while the highest-resource cells improve almost to the end.

4.4 Connector training versus off-the-shelf models

Table 4 compares each trained LisTaya checkpoint against the strongest off-the-shelf model we evaluate on the same test set. In every cell evaluated

so far, training a projector on a few hours of in-language audio beats the best downloadable model, and the margin is largest exactly where the baseline is weakest.

4.5 A language neither component supports

Kreol Seselwa is supported by neither the Whisper encoder nor any Tiny Aya decoder, and it is the one cell where both frozen components operate outside their coverage. The four decoders span 17.92–21.35 CER on it, a range of 3.43 CER. The projector therefore learns a usable alignment for a language absent from both pretraining corpora, and which regional variant it is paired with matters as little there as elsewhere.

Limitations

Most cells are trained once. Ten of the 52 runs have a replicate, which is what lets us state a between-run standard deviation of 1.17 CER, but the remaining cells carry no direct uncertainty estimate, and the design’s minimum detectable effect of 4.90 CER is well above the effects that would be practically interesting. A negative result at this width bounds the effect rather than excluding it.

Ten of the twelve languages train on WorldSpeech and evaluate on FLEURS, so most reported numbers combine decoder choice with a domain shift from formal recorded speech to read speech. We report in-domain WorldSpeech test results alongside them where available, but the two are not separable in every cell.

The study measures transcription accuracy only. Whether decoder specialization affects downstream spoken-language understanding, where the decoder’s own multilingual knowledge is more directly exercised, is not addressed here.

Finally, the comparison assumes the Tiny Aya variants differ only in post-training data. We take that from the model family’s own documentation rather than verifying it independently.

Acknowledgments

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Language	Region	Match	Mism.	Global	Δ
Amharic	Earth	29.25	27.93	27.00	1.32
Tamil	Fire	43.63	58.33	61.21	-14.70
Urdu	Fire	22.24	20.81	17.64	1.44
Hausa	Earth	33.43	34.62	35.03	-1.19
Marathi	Fire	13.57	14.01	14.14	-0.44
English	Water	12.05	4.67	4.96	7.39
Spanish (LatAm)	Water	3.41	3.35	3.40	0.07
French	Water	6.36	6.79	6.61	-0.43
Hindi	Fire	13.11	12.05	13.18	1.06
Indonesian	Water	5.74	6.14	6.05	-0.40
Swahili	Earth	13.53	13.68	12.60	-0.15

Table 2: Best CER per language for the region-matched decoder, the mean of the two mismatched regional decoders, and LisTaya-Global. Δ is matched minus mismatched; negative favours the matched decoder. Ordered by resource tier.

Tier	Langs	Eval-loss rise	Frac. to best
very low	2	0.211	0.282
low	1	0.188	0.292
mid	2	0.041	0.627
high	7	0.003	0.816

Table 3: Overfitting by resource tier. Both columns are medians of per-language medians, so a tier with seven languages cannot outweigh one with two.

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Language	Test set	Domain	Base.	LisTAya	Δ
Hausa	ha_ng	FLEURS	95.56	68.72	-26.84
Swahili	sw_ke	FLEURS	51.52	35.02	-16.50
Swahili	sw_ke	in-domain	25.93	20.85	-5.08
Swahili	sw_tz	in-domain	78.10	25.27	-52.83

Table 4: WER of the best off-the-shelf baseline against the best LisTAya checkpoint on the same test set. Negative Δ favours the trained projector.

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A Example Appendix

This is an appendix.